Design and Implementation of an Intelligent Security Management Platform for Private Clouds: A Two-Phase Approach

Yang Zhou, Lingxiao Zhou, Taewoon Kim*

Pusan National University

{zhouyang, zlx202488581, *taewoon}@pusan.ac.kr

프라이빗 클라우드 환경을 위한 지능형 보안 관리 플랫폼의 설계 및 구현: 2 단계 접근법

저우양, 조우 링샤오, 김태운*
부산대학교

Abstract

The cloud-native private cloud environments have become increasingly complex, introducing new cybersecurity challenges. Traditional rule-based security systems and signature libraries are mainly effective against known attacks and have limited capability in detecting unknown threats. In addition, the lack of labeled attack data restricts the application of supervised learning methods in cloud security. To address these issues, this paper presents the design and implementation of an intelligent security management platform for private clouds based on a two-phase approach: A) an off-line phase for Infrastructure as Code (IaC) validation, and B) an on-line phase for real-time anomaly detection. In the off-line phase, the IaC validation improves the consistency and security of the cloud resource to deploy. In the on-line phase, the proposed platform adopts eBPF-based real-time traffic monitoring, One-Class SVM-based anomaly detection, and Zero-Shot Learning to improve the detection of both known and unknown attacks while reducing CPU overhead. Overall, the proposed platform improves automated management and real-time security protection capabilities in cloud-native private cloud environments.

Ⅰ. Introduction

With the widespread adoption of Kubernetes and Docker, cloud-native architectures have become a core component of modern private cloud environments. Although these environments provide high scalability and automated resource management, increasingly complex network topologies also introduce new cybersecurity challenges. Traditional rule-based security systems [1] and signature libraries are mainly effective against known attacks and have limited capability in detecting unknown threats. In addition, the lack of labeled attack data restricts the application of supervised learning methods in cloud security. Meanwhile, Infrastructure as Code (IaC) technologies such as Terraform [2] are widely used for automated Kubernetes management, but configuration errors may also introduce security risks. Therefore, modern private cloud environments require both secure resource deployment and intelligent real-time threat detection.

To address these challenges, this paper presents the design and implementation of an intelligent security management platform for private clouds based on a two-phase approach: A) an off-line phase for Infrastructure as Code (IaC) validation, and B) an on-line phase for real-time anomaly detection. The proposed platform integrates eBPF-based real-time traffic monitoring, Zero-Positive Deep Learning, and Zero-Shot Learning to improve real-time security protection in cloud-native environments.

Ⅱ. Method

Our proposed platform consists of two main phases: A) an off-line phase for IaC validation based on Terraform, and B) an on-line phase for real-time anomaly detection. The overall architecture is illustrated in Fig 1.

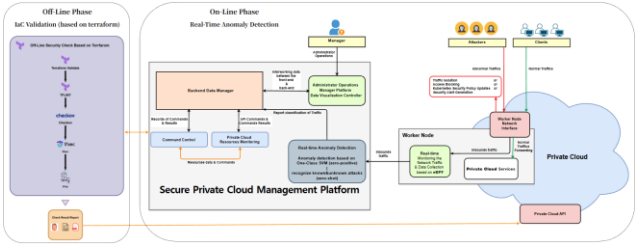


Fig 1. Architecture of the proposed intelligent private cloud security management platform.

A. Off-Line Phase: IaC Validation

In the off-line phase, the platform utilizes Infrastructure as Code (IaC) technology [2] to automate the deployment and management of Kubernetes resources and services. Before deployment, IaC validation tools are employed to verify Kubernetes configurations and security policies, as shown in Fig 2, thereby reducing potential security risks caused by misconfigurations. In addition, the platform provides a graphical user interface for Terraform operations and Kubernetes cluster monitoring, enabling automated and secure cloud resource management.

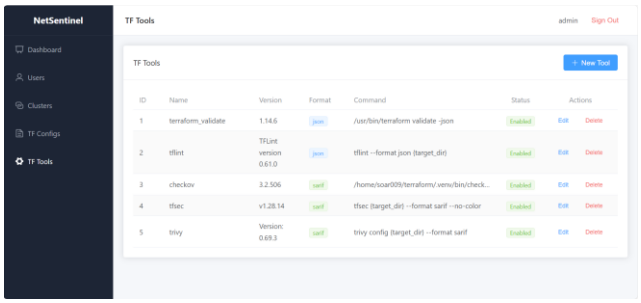


Fig 2. Web interface of Terraform-based IaC validation tools for Kubernetes security configuration and policy verification.

B. On-Line Phase: Real-Time Anomaly Detection

In the on-line phase, the platform continuously collects real-time network traffic and system status information through an eBPF-based traffic collection mechanism. Since fully implementing true zero-copy transmission is difficult in cloud-native environments due to Linux kernel operations, container networking, and virtualization complexity, eBPF is adopted to reduce CPU overhead, improve traffic collection efficiency, and minimize transmission latency. The collected traffic and anomaly detection results are visualized through a dynamic dashboard for real-time monitoring.

For anomaly detection, this study employs a zero-positive deep learning approach based on One-Class SVM model [3] to learn normal traffic patterns and identify anomalous behaviors. Furthermore, a Zero-Shot Learning mechanism is introduced to recognize unknown attacks by measuring the similarity between real-time traffic and predefined attack semantic prototypes. Once abnormal traffic is detected, the system automatically performs security responses such as traffic isolation, access blocking, Kubernetes security policy updates, and security alert generation, thereby improving security response efficiency and reducing manual intervention.

Ⅲ. Conclusion

This paper presented the design and implementation of an intelligent security management platform for private clouds based on a two-phase approach: A) an off-line phase for Infrastructure as Code (IaC) validation, and B) an on-line phase for real-time anomaly detection. The platform integrates eBPF-based traffic monitoring, One-Class SVM-based anomaly detection, and Zero-Shot Learning to improve the detection of both known and unknown attacks in cloud-native environments. In the off-line phase, the IaC validation enhances the consistency and security of Kubernetes resource deployment. In the on-line phase, the eBPF-based traffic collection mechanism improves real-time traffic analysis efficiency while reducing CPU overhead.

Overall, the proposed platform improves real-time security protection and automated management capabilities in private cloud environments. Future work will further conduct experimental analysis and optimization studies on system performance, detection accuracy, and operational stability in large-scale cloud network environments.

ACKNOWLEDGMENT

This work was partly supported by the IITP(Institute of Information & Communications Technology Planning & Evaluation)-ITRC(Information Technology Research Center) grant funded by the Korea government(Ministry of Science and ICT)(IITP-2026-RS-2023-00259967, 50%) and 「Fostering Project of R&D promotion complex in Busan」 funded by the Korea government(MSIT) and a local government(Busan)(2710089900, 50%).

REFERENCES

[1] V. R. Kommidi, S. Padakanti, and V. Pendyala, "Securing the cloud: A comprehensive analysis of data protection and regulatory compliance in rule-based eligibility systems," *Int. J. Res. Comput. Appl. Inf. Technol. (IJRCAIT)*, vol. 7, no. 2, pp. 432- 447, Jul.- Dec. 2024, doi: 10.5281/zenodo.13991239.

[2] Y. Brikman, *Terraform: Up and Running: Writing Infrastructure as Code*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.

[3] G. S. Fuhnwi, V. Adedoyin, and J. O. Agbaje, "An Empirical Internet Protocol Network Intrusion Detection using Isolation Forest and One-Class Support Vector Machines," *Int. J. Adv. Comput. Sci. Appl. (IJACSA)*, vol. 14, no. 8, pp. 1- 8, 2023.